A One-Dimensional Angular Enrichment Test on the UNESCO World Heritage Longitude Distribution: Domed Monuments, the Babylonian Beru, and a Reproducible Spatial-Pattern Template

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Code and data: `github.com/SethTenenbaum/gerizim-paper-a`

April 2026
v1.0.9

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Abstract

We present a transferable quantitative template for testing candidate angular units against externally-curated monument corpora, and apply it to the UNESCO World Heritage Cultural and Mixed site list ($N = 1011$). The template uses a one-dimensional angular enrichment test against a fixed geometric null (4%), a pre-registered morphological keyword classifier, and a fully reproducible open-source pipeline.

Applied to the 0.1-beru interval (3°, the smallest integer sub-division of the Babylonian angular unit; MUL.APIN, c. 1000 BCE), the method detects a global 3° periodic structure in the UNESCO longitude distribution. The primary confirmatory tests ($k = 3$, Bonferroni-corrected) find: domed and spherical monuments ($N = 90$) show 12.2% Tier-A+ rate against the 4% geometric null ($Z = 3.21$, $p_{\text{adj}} = 0.003$ ***); and A+-bearing harmonics host 4.19× more total sites than non-A+-bearing harmonics ($p_{\text{adj}} = 0.01$ **). Leave-one-out checks confirm no single site drives either result. The Americas sub-corpus shows directional depletion at the same harmonics, consistent with an Eurasian-anchored grid.

This study began with Mount Gerizim (35.27°E) as the historical anchor; the phase coincidence with the optimal band introduces a circularity (§1.1) that limits all findings to exploratory status. The principal contribution is the methodological template itself: a fixed external corpus, a geometric null with no free parameters, and an archived pipeline applicable to any candidate angular unit.

Keywords: longitude periodicity, monument distribution, UNESCO World Heritage, Babylonian beru, ancient units of measurement, domed architecture, stupa, spatial statistics, binomial enrichment test, heritage site geography, archaeometry, Levantine metrology, quantitative archaeology, Mount Gerizim

1 Introduction

This paper applies a one-dimensional angular enrichment test to the UNESCO World Heritage Cultural and Mixed corpus, an externally-curated and publicly fixed dataset, using a reproducible computational pipeline deposited in full at https://github.com/SethTenenbaum/gerizim-paper-a. The methodological approach relies on a fixed external corpus, a geometric null with no free parameters, and an open archival pipeline. It is intended as a transferable template for spatial pattern-detection studies in archaeological heritage data, applicable to any candidate angular unit and corpus.

The corpus ($N = 1011$) appears to contain a global 3° periodic structure in its longitude distribution. Every longitude of the form x.18°E produces the maximum observed Tier-A+ count when used as an angular reference (sites within $\lesssim$ 6.7 km of a harmonic node, against a 4% geometric null). The 3° spacing equals the smallest integer sub-division of the Babylonian *beru* unit (0.1 beru = 3° = 3 *uš*, MUL.APIN, c. 1000 BCE). The analytic consequences of the Gerizim phase-coincidence that initiated this study are defined in §1.1.

1.1 Circularity Containment Protocol

This study began with Mount Gerizim (35.269°E) as the historical anchor for a beru-unit test on the UNESCO corpus. A global phase sweep subsequently revealed that Gerizim's longitude falls within the x.18°E optimal band, creating a circularity: reporting the Levantine corridor statistics was conditioned on observing the very phase alignment those statistics describe. This identification problem cannot be closed by any statistical method applied to this corpus. Accordingly, all results are treated as *exploratory* without exception, the Bonferroni correction serves as a descriptive bound on effect magnitude, and the Levantine statistics are the motivated entry point rather than independent evidence.

The circularity contaminates the Levantine corridor statistics (Tests 1–3 as computed from Gerizim) and the x.18°E framing. It does *not* contaminate the existence of the 3° periodic structure (recoverable by any unanchored 120-phase scan), the dome enrichment (Test 2, all 120 anchors produce identical A+ counts for the dome population), the cluster asymmetry (Test 3, a full-corpus result requiring no keyword filter), or the unit sensitivity sweep. The full canonical treatment, including the *forking paths* analysis (Gelman & Loken, 2014) and the impossibility of a purely spatial resolution, is given in §6.1.

Why this design addresses key methodological criticisms. Prior metrological investigations were rightly criticized for investigator-assembled site lists (Williamson & Bellamy, 1983; Freeman, 1976; Kendall, 1974). Three features distinguish this study: (1) the UNESCO site list was compiled without reference to this hypothesis and is fixed; (2) the 0.1-beru interval is documented in cuneiform sources (Hunger & Steele, 2019) as the finest integer subdivision, limiting the parameter space; (3) tier boundaries were fixed on geometric grounds and keywords are simple morphological matches.

Positioning within quantitative archaeometry. This paper belongs to the tradition of probability-based spatial analysis in archaeometry and archaeoastronomy (Magli, 2013, 2016; Ruggles, 1999, 2015; Silva, 2020). The use of a fixed external corpus, a geometric null free of fitted parameters, and a publicly archived computational pipeline responds directly to the methodological standards articulated in those frameworks and in the ICOMOS-IAU thematic study on World Heritage archaeoastronomy (Ruggles & Cotte, 2010). The design is applicable to any candidate angular unit and any externally

curated monument corpus, making the template reusable independently of the specific metrological hypothesis tested here.

Structure of the paper. Section 2 provides background on the beru unit. Section 3 describes data, classification, and multiple-comparisons protocol. Section 4 presents each test's design, result, and robustness together. Section 5 reports additional robustness checks. Section 6 catalogs limitations, including the full circularity treatment (§6.1). Section 7 evaluates alternatives and concludes. The Appendix contains the keyword audit.

2 Background

2.1 The Beru Unit

The *beru* (30° of arc) is attested in the MUL.APIN cuneiform compendium (Hunger & Steele, 2019; Watson & Horowitz, 2011), dated on broad grounds to the late second or early first millennium BCE. In this system the unit divided the ecliptic into twelve equal segments and served for both stellar and terrestrial angular measurement. The terrestrial use is attested explicitly in MUL.APIN II ii 43–iii 15, which converts beru of daylight into ground distances (Hunger & Steele, 2019, pp. 2–15). The sub-division into 30 *uš* (1 uš = 1°) is well attested (Hunger & Steele, 2019; Robson, 2008), and the 0.1-beru interval (3°) tested here corresponds to exactly 3 *uš*.

The beru also functioned as a unit of territorial description. Royal inscriptions record inter-regional distances in beru between major cities (Neugebauer, 1975; Robson, 2008). On the Babylonian *Imago Mundi*, distances between the "islands" beyond the encircling Salt Sea are marked as multiples of beru. The *kudurru* (boundary stone) tradition records territorial extents in cuneiform, and many kudurru were deposited in temples as legally authoritative records (Robson, 2008; Powell, 1987).

Longitudinal positioning at beru-scale precision did not require a dedicated geodetic instrument. Systematic meridian transit observation and comparative eclipse timing, both well attested in the cuneiform astronomical record (Hunger & Steele, 2019; Neugebauer, 1975), are sufficient to derive inter-site angular separations at 1° resolution or better. The gnomon, attested across Mesopotamian, Egyptian, and Indian traditions (Neugebauer, 1975; Shamasastry, 1915), provides an independent means of establishing true meridian. The absence of a recovered instrument calibrated explicitly in beru subdivisions is therefore not equivalent to the absence of the operational capacity to execute positional measurements at that scale.

This combination of a fractional angular unit, temple-deposited territorial records, and attested observational infrastructure provides the documentary basis for testing whether a 0.1-beru harmonic spacing is detectable in the longitude distribution of major monuments. This study tests the statistical pattern only. The mechanism by which any such correspondence could arise is not documented and is outside the scope of this paper.

2.2 Ancient Metrology and Geodetic Practice

Whether pre-modern societies employed standardized angular measures across large geographic scales has a long and contested history. Thom's "megalithic yard" (Thom, 1967, 1971, 1964) stimulated statistical refinement (Kendall, 1974) and methodological critique (Freeman, 1976). Powell's treatment of Mesopotamian metrology (Powell, 1987) remains the standard reference. More recently, Magli (Magli, 2013, 2016) has applied Monte

Carlo methods to archaeoastronomical alignments, and Ruggles and Silva have advanced probability-based frameworks for evaluating orientation claims (Ruggles, 1999, 2015; Silva, 2020). The ICOMOS-IAU thematic study (Ruggles & Cotte, 2010) provides institutional context for archaeoastronomical analysis of World Heritage properties. Medieval cartographic traditions also employed directional reference systems anchored to sacred geography, with Jerusalem and Mecca serving as orienting meridians in mappamundi and qibla charts (Harvey, 1987), providing historical context for the Levantine corridor hypothesis.

This study departs from prior work in three respects. First, the site list is externally curated (the UNESCO World Heritage List), eliminating selection bias. Second, the test is one-dimensional (longitude only), yielding a well-defined geometric null (4%) with no free parameters. Third, the analysis is fully computational and publicly archived (Tenenbaum, 2026).

2.3 The Anchor: Mount Gerizim

The circularity arising from Gerizim's longitude coinciding with the optimal phase band is defined in §1.1.

Mount Gerizim (35.269°E) is the primary reference anchor because of its documented *tabbur ha-aretz* ("navel of the earth") role in Samaritan cosmographic tradition (Purvis, 1968; MacDonald, 1964; Crown, 1989), analogous to Babylon's meridian in cuneiform astronomical records (Hunger & Steele, 2019). As a prominent mountain (881 m) in the central highlands, it would have served as a natural survey reference point. No optimization across alternative Levantine anchors was performed.

Gerizim's tentative-list coordinate (ref. 5706, DMS E35 16 08) is externally sourced and does not appear in the UNESCO inscribed corpus.[1] Jerusalem (35.2317°E), approximately 4.1 km east of Gerizim, is an inscribed corpus member. It is self-excluded when used as anchor and produces the same A+ count (§5.3). The enrichment belongs to the 4.1 km corridor, not to the specific choice of anchor within it. Mecca reaches only $p \approx 0.057$ (ns); Mt. Meru (Tanzania, 36.750°E) reaches $p \approx 0.305$ (ns) and Mt. Kailash, the most cited geographic candidate for the cosmological axis mundi, reaches $p \approx 0.206$ (ns). Neither alternative anchor approaches significance.

2.4 Interpretive Framework

This paper is a pattern-detection study, not a historical attribution. Three classes of explanation are evaluated throughout.

1. **Geographic concentration.** Major civilizations that produced UNESCO-ranked monuments developed in Eurasian longitude bands that align with 0.1-beru harmonics. The dedicated geographic-concentration simulation (§4.1) shows this remains a viable first-level account.

2. **Metrological transmission.** A 3° (0.1-beru) angular reference system anchored near the Levant informed the placement of major foundations. No mechanism is

[1] UNESCO World Heritage Centre, *Mount Gerizim and the Samaritans*, Tentative List ref. 5706 (State of Palestine, submitted 02 April 2012; criteria (iii)(vi); coordinates N 32°12'44" E 35°16'08"). https://whc.unesco.org/en/tentativelists/5706/. Archived rendered copy: https://ia601602.us.archive.org/33/items/unesco_5706_rendered/unesco_5706_rendered.html.

documented.

3. **Unmodeled statistical artifact.** A systematic bias in UNESCO curation, geocoding, or the discrete longitude distribution produces the apparent periodicity. An artifact account must additionally explain the dome enrichment and the cluster asymmetry.

Adjudicating between explanations 1 and 2 requires independent corpora.

3 Data and Methods

3.1 Corpus Construction

The UNESCO World Heritage XML dataset contains 1248 inscribed nominations across all categories. Two outcome-blind filters yield the analysis corpus: (1) Natural-only sites are excluded (-235, retaining Cultural and Mixed); (2) entries lacking geocoordinates are excluded (-2). The result is $N = 1011$ sites with valid longitudes.

3.2 Geodetic Framework

The reference anchor is Mount Gerizim at 35.269°E, from Palestine's UNESCO tentative list (ref. 5706, 2012).[2] Gerizim is not on the inscribed World Heritage List and does not appear in the test-site corpus. Jerusalem is also excluded in both sweeps (§2.3).

For a site at longitude λ, the beru-distance is $B = |\lambda - 35.269|/30$ and the beru-deviation from the nearest 0.1-beru harmonic is $\delta = |B - \text{round}(B, 0.1)|$. The tier classification is a fixed angular criterion. Kilometer labels (e.g. $\lesssim 6.7$ km for Tier-A+) are equatorial approximations computed as $\delta \times 30 \times 111$ km/°, becoming smaller at higher latitudes. This makes the angular criterion *conservative* at high latitudes. A geodesic sensitivity check confirms that applying a strict physical threshold would increase the A+ count to 70, strengthening all primary results. No latitude correction is applied.

The geometric null for Tier-A+ is $2 \times 0.002/0.1 = 4\%$, assuming uniform longitudes. This serves as a convenient reference, but the primary tests are simulation-based null models (§5.5) that preserve the empirical longitude distribution. A Gaussian KDE produces 40.4 ± 6.2 expected A+ ($\sim$4.0%), confirming the geometric null is not materially inflated by longitude clumping.

3.3 Metrological Basis for the 0.1-Beru Spacing

The harmonic spacing tested is 0.1 beru (3°), the smallest integer sub-division of the beru attested in cuneiform metrology, equalling exactly 3 *uš* (Hunger & Steele, 2019; Robson, 2008; Powell, 1987). It was selected because it is the finest metrologically attested integer sub-division that yields a tractable test (120 harmonics, 4% geometric null). Coarser spacings (0.5 beru, 1 beru) are uninformative; finer sub-divisions (0.02 beru) do not produce the signal (Table 8). No spacing was selected post-hoc to maximize the result.

The 0.05-beru spacing (6°) also reaches significance (Table 8). This is structurally expected because every 0.10-beru harmonic is also a 0.05-beru harmonic, making the 0.05-beru result a coarser sub-sampling of the same periodicity, not a separate signal.

[2] See footnote in §2.3 for full citation and archived URL.

3.4 Tier Classification

Sites are classified by δ, the angular deviation from the nearest 0.1-beru harmonic. All tier boundaries and test populations were fixed on metrological grounds, not tuned to the data.

- **Tier A++**: $\delta \leq 0.0002$ beru ($\lesssim 0.67$ km). $P_0 = 0.4\%$. Descriptive only, not tested.
- **Tier A+**: $\delta \leq 0.002$ beru ($\lesssim 6.7$ km). $P_0 = 4\%$. Primary test threshold.[3]
- **Tier A**: $\delta \leq 0.010$ beru ($\lesssim 33$ km). $P_0 = 20\%$.
- **Tier B**: $\delta \leq 0.050$ beru.

3.5 Test Populations

All keyword-based populations are approximate. The objective is to reduce false positives as far as possible while accepting that a simple morphological keyword match cannot eliminate interpretive ambiguity entirely. The primary (raw) sweep deliberately tolerates known false negatives for simplicity; the context-validated variant (Test 2x) applies sentence-level gating to reduce false positives further, at the cost of a more complex classification rule. Both populations are reported; the raw sweep serves as the conservative primary test because its inclusion criterion is fully mechanical.

Test 2 (dome/spherical, primary). 7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text ($N = 90$). This raw sweep is over-inclusive: it admits sites where the keyword refers to geological or vernacular features rather than monumental architecture. The context-validated population ($N = 83$) is reported alongside all primary results; it yields stronger significance, consistent with the rejected sites being longitude-diluting false positives.

Test 2b (mound evolution, sensitivity variant). Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun, mound*). Not counted separately in the Bonferroni family.

Test 3 (cluster asymmetry, primary). Full corpus ($N = 1011$), no keyword filter.

Test 4 (temporal gradient, descriptive). Cochran-Armitage trend test on the full corpus, no keyword filter.

Test E (founding classifier, post-hoc). A context-aware keyword classifier (`lib/founding_filter.py`) iteratively tuned after data inspection to reduce false positives while accepting residual interpretive ambiguity (Appendix A.2). The classifier is fully exploratory: it addresses what kinds of sites beyond domed monuments tend to fall on harmonic nodes, but carries no weight in the primary evidential chain.

3.6 Multiple-Comparisons Protocol

Primary test family ($k = 3$).

- **Test 1** (full corpus, $N = 1011$): global enrichment above 4% null (descriptive backdrop).

[3] $2 \times 0.002/0.1 = 4\%$. Km labels are equatorial approximations.

- **Test 2** (domed/spherical monuments, $N = 90$): morphological sub-population enriched above 4% null.
- **Test 3** (cluster asymmetry, full corpus, no keyword filter): A+-bearing harmonics host more total sites.

Tests 2 and 3 address structurally different questions on different populations and carry the primary evidential weight. All three pass the Bonferroni bound at $k = 3$ (dome $p_{\text{adj}} = 0.003$ ***, cluster $p_{\text{adj}} = 0.01$ **).

Other tests. Test 4 (temporal gradient, Cochran-Armitage) is reported as descriptive context; it was demoted from the primary family because the temporal trend is substantially explained by regional composition (§4.4), and removing it from the family strengthens the Bonferroni bound on the two primary results. Test 2b is a sensitivity variant of Test 2. Test E (founding classifier) is fully post-hoc (Appendix A.2). Test B (Rayleigh) is a structural description (§4.3) and is not included in either table.

Summary table.

Table 1: Multiple-comparisons summary. Primary test family: Tests 1–3 ($k = 3$, $\alpha = 0.05/3$). All results are exploratory (§1.1).

Test	N	Raw p	p_{adj}, k=3	Status
Test 2 (dome A+)	90	0.0009^a	0.003 ***	**Primary**
Test 3 (cluster asym.)	1011	0.0033^b	0.01 **	**Primary**
Test 1 (full corpus)	1011	0.0103^a	0.031 *	Exploratory
Test 4 (temporal)		0.0481^b	—	Exploratory‡
Test 2b (mound evol.)	132	0.0009^a	N/A	Sensitivity
Test E (founding)		^{a}	N/A	Post-hoc
Test B (Rayleigh)†	56	0.0000^b	N/A	Structural

^{a}Exact binomial p (one-tailed vs. geometric null); ^{b}permutation p.
†Test B: $R \approx 1$ by A+ construction; excluded from all family counts (§4.3).
‡Test 4 demoted to exploratory; not in k=3 family; no adj-p computed (§3.6).

A comprehensive FDR audit (34 tests, Benjamini & Hochberg 1995) retains 21 at $q < 0.05$, of which 6 survive Bonferroni at $k = 34$ (Table 2).

Table 2: Full FDR audit across all 34 reported tests, sorted by raw p-value. BH q-values: Benjamini & Hochberg (1995). Test B excluded from family counts. All results are exploratory (§1.1).

Test	N	Raw p	BH q (k=34)	Bonf. p_{adj} (k=34)	Status
Test B (Rayleigh)†	56	0.0000	N/A	N/A	Structural†
Test 2 (dome A+)	90	0.0009	0.0054	0.0321	**Primary**
Test 3 (cluster asym.)	1011	0.0033	0.014	0.1122	**Primary**
Test 2b (mound evol.)	132	0.0009	N/A	N/A	Sensitivity
Test 1 (full corpus)	1011	0.0103	0.0219	0.3502	Exploratory
Test 4 (temporal)		0.0481	—	—	Exploratory
Test E (founding)			N/A	N/A	Post-hoc

21 tests survive BH-FDR ($q < 0.05$); 6 survive Bonferroni ($k = 34$).
†Test B excluded: $R \approx 1$ by A+ definition.

3.7 Null Models

Three simulation approaches (code: `simulation_null_model.py`, 100,000 trials each) and two dedicated geographic-concentration nulls are described with their results in §5.5. An independent Wikidata P1435 control corpus ($N = 62{,}785$) is used as external reference (§6).

4 Results

Each subsection below describes one test's design and result together. Test E (founding classifier, post-hoc) is in Appendix A.2.

4.1 Domed and Spherical Monuments (Test 2, Primary)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text. All keywords are treated identically (positive whole-word match, no context gating), making the population mechanically determined by the keyword list. Final population: $N = 90$. Permutation $Z = 3.21$, $p = 0.004$; bootstrap $p = 0.006$.

Table 3: UNESCO domed/spherical monument test ($N = 90$, raw keyword sweep). Primary evidence against the chance explanation: simulation permutation $Z = 3.21$, $p = 0.004$ (§5.5). Exact binomial results are reference baselines assuming uniform longitude spacing. The non-significant χ^2 is discussed in the text.

Metric	Obs	Exp(H_0)	Rate [95% CI]	Enrich.	p	
Tier-A+ ($\leq$ 0.002 beru)	11	3.6	12.2% [6.3, 20.8]	3.06×	0.0009	***
Tier-A ($\leq$ 0.010 beru)	29	18.0	32.2% [22.8, 42.9]	1.61×	0.0043	**
χ^2-uniform (5 bins, df = 4)	—	—	—	—	0.0732	ns

Expected under H_0: A+ = 4%, A = 20%. CIs are Clopper-Pearson exact.
Tier-B: 61 / 90 Mean δ: 0.0220 beru
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$ (trend-level), ns = not significant.

Tier-A+ sites ($N = 11$, 12.2%, 3.06×, permutation $p = 0.004$, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$): Khoja Ahmed Yasawi (0.2 km), Boyana Church (0.3 km), Lumbini (0.8 km), Humayun's Tomb (2.0 km), Pyu Ancient Cities (2.3 km), Silk Roads: Chang'an-Tianshan (2.5 km), Trulli of Alberobello (3.6 km), Old City of Jerusalem (4.1 km), Borobudur (4.3 km; Miksic 1990), Kizhi Pogost (5.0 km), Echmiatsin/Zvartnots (6.2 km).

A leave-one-out check confirms no single site drives the result: removing any A+ site yields $N = 89$, A+= 10, $p = 0.0030$. Leave-2-out (worst case across all $\binom{11}{2}$ combinations): $p = 0.0087$. Leave-3-out (worst case): $p = 0.0233$. Code: `dome_leave_k_out.py`.

Context-validated robustness check (Test 2x). Applying sentence-level architectural confirmation reduces the population to $N = 83$ (7 non-architectural false positives rejected; full list in Appendix A.1). Test 2x yields $p = 0.0005$, 3.31× enrichment, stronger than the raw sweep, consistent with rejected sites diluting the signal. Because validation rules were developed after the initial sweep, Test 2x is excluded from the primary Bonferroni family.

Geographic-concentration null (dedicated simulation). Domed monuments are concentrated in the Eurasian corridor: 77% fall in 20°–120°E versus 41% of the full corpus. Two targeted null models test whether this explains the dome enrichment (code: `dome_geographic_concentration_test.py`, 100,000 trials).

Null A (within-dome bootstrap): Draw $N = 83$ longitudes from the dome sites' own list. Mean = 11.01 A+, $p = 0.5514$ (ns). The dome sites' positions are concentrated near harmonic longitudes.

Null C (restricted geographic draw): Pool all 881 full-corpus sites within ±5° of any dome site. Pool mean = 4.81 A+ (5.8%), $p = 0.0086$ (**). Dome sites (11 A+, 13.3%) significantly exceed the broader footprint. Sensitivity at ±2° and ±10° yields consistent results.

Reconciliation. Null A confirms that the dome sites' longitude distribution reproduces their enrichment (geographic concentration is necessary). Null C establishes that other monuments at the same longitudes are significantly less enriched (geographic concentration is not sufficient). The residual question is whether the sub-longitude precision of domed monument placement reflects metrological intent or an unmodeled aspect of these civilizations. Adjudicating requires independent corpora.

In the global anchor sweep (§5.3), Gerizim scores 55 A+ (98.3th percentile); Jerusalem scores 56 (98.3th). The meta-keyword test confirms that only % of candidate words achieve comparable enrichment.

4.2 Hemispherical Mound Evolution (Test 2b [Sensitivity Variant])

Extends Test 2 by adding earthen mound-stage keywords (*tumulus, tumuli, barrow, barrows, kofun, mound*); the combined population ($N = 132$) comprises 42 mound-stage sites added to the core domed/spherical population ($N = 90$), plus 6 sites matching both keywords. This test is a sensitivity variant of Test 2, not an independent hypothesis, and is not counted in the Bonferroni family. Mound-only sites show no significant enrichment (6.2%, $p = 0.3012$, ns).

Table 4: Test 2b: hemispherical mound evolution ($N = 132$). Combined tumulus → stupa → dome population.

Metric	Obs	Exp(H_0)	Rate	p	
Tier-A+ (≤ 0.002 beru, ≤ 6.7 km)	14	4.16	10.6%	0.0009	***
Tier-A (≤ 0.010 beru, ≤ 33 km)	41	20.80	31.1%	0.0017	**

Expected under H_0: A+ = 4%, A = 20%. Enrichment: A+ = 2.65×, A = 1.55×.
*** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, † $p < 0.10$, ns = not significant.

Stage-by-stage breakdown (Table 5):

Table 5: Tier-A+ enrichment by evolutionary stage. Null rate $P_0 = 4\%$.

Stage	N	A+	Rate	Enrichment	p (binomial)
Mound (tumulus/barrow/kofun)	48	3	6.2%	1.55×	0.3012 ns
Stupa (stupa/stupas)	18	4	22.2%	5.56×	0.0050 **
Dome (dome/tholos/spherical)	75	8	10.7%	2.68×	0.0102 *
Combined	**132**	**14**	**10.6%**	**2.65×**	**0.0009 *** **

Stage N values sum to more than the combined total because 6 sites match keywords from multiple stages and are counted once in the combined row. A+ counts by stage are computed within each stage population independently; a site matching two stages contributes its A+ status to each stage separately, so stage A+ values may not sum to the combined A+ count. The combined A+ total is the count of unique A+ sites across the full Test 2b population ($N = 132$).

The combined population ($N = 132$, A+ = 10.6%, $p = 0.0009$ ***) is consistent with Test 2. The mound stage does not reach significance (6.2%, $p = 0.3012$, ns). The signal is carried by the stupa (5.56×, $p = 0.0050$ **) and dome sub-stages. Leave-one-out checks confirm stability for all sub-stages.

Stupa geographic-concentration null. Stupa sites are concentrated in South/Southeast Asia (72% in 70°–110°E). Null A (within-stupa bootstrap): $p = 0.5920$ (ns). Null C (restricted ±5° draw): $p = 0.0196$ (*). The structure parallels the dome result: geographic proximity to harmonic longitudes is necessary but not sufficient to explain the morphological specificity.

4.3 Cluster Asymmetry (Test 3, Primary)

Full corpus ($N = 1011$), no keyword filter. For each site, count other UNESCO sites within ±0.3° longitude; compare cluster sizes for A+ vs. non-A+ sites (Mann-Whitney

U, permutation test). At the harmonic level, compare per-harmonic site counts for A+-bearing vs. non-A+ harmonics.

Table 6: Cluster asymmetry: A+ vs. non-A+ sites in the full UNESCO corpus ($N = 1011$; cluster radius $\pm 0.3°$ lon)

Metric	A+ sites	Non-A+ sites	Ratio / Z	p
Mean cluster size (sites within $\pm 0.3°$ lon)	6.77	5.08	1.33×	
Mann-Whitney U (one-tailed)				0.0182 *
Permutation (10,000 shuffles)			$Z = 2.82$	0.0033 **
Harmonic-level density (all Tier $\leq$ B sites per 0.1-beru harmonic):				
Harmonics with ≥ 1 A+ site (28)	mean = 29.4 sites			
Harmonics with 0 A+ sites (27)	mean = 7.0 sites		4.19×	< 0.0001 ***

A+-bearing harmonics host 4.19× more sites on average than non-A+ harmonics ($p < 0.0001$). Sites at harmonic nodes cluster more densely than sites at harmonic midpoints (9.85 vs. 5.83 neighbors; 1.69×, $p = 0.0238$).

Subtest: does clustering produce the unit-sensitivity peak? A joint permutation test (50000 iterations, two independent shuffles per trial) asks whether extreme clustering mechanically produces the canonical-unit peak. Of 182 clustering-matched permutations ($p_{\text{cluster}} = 0.0036$), the conditional fraction producing the peak is 0.1209, nearly identical to the unconditional rate of 0.1681. The unit-sensitivity peak and the clustering signal are statistically independent; the primary evidence remains the clustering (Test 3, $p = 0.0033$) and dome enrichment (Test 2, $p = 0.004$).

Methodological note: Rayleigh phase-lock (Test B). The near-perfect Rayleigh R of the A+ sub-sample is a mathematical consequence of the A+ classification criterion (§1.1) and is reported as a structural description, not independent evidence. A permutation check confirms that a random draw of 56 longitudes from the full corpus does not reproduce the observed phase concentration ($R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} < 0.0001$, 10,000 shuffles, seed 42). The full corpus Rayleigh $R = 0.0224$ ($p_{\text{perm}} = 0.3715$, ns) confirms no spurious global periodicity in the raw longitude file. Full details: `x18_periodicity_formal_test.py`.

4.4 Global Enrichment (Test 1, Descriptive)

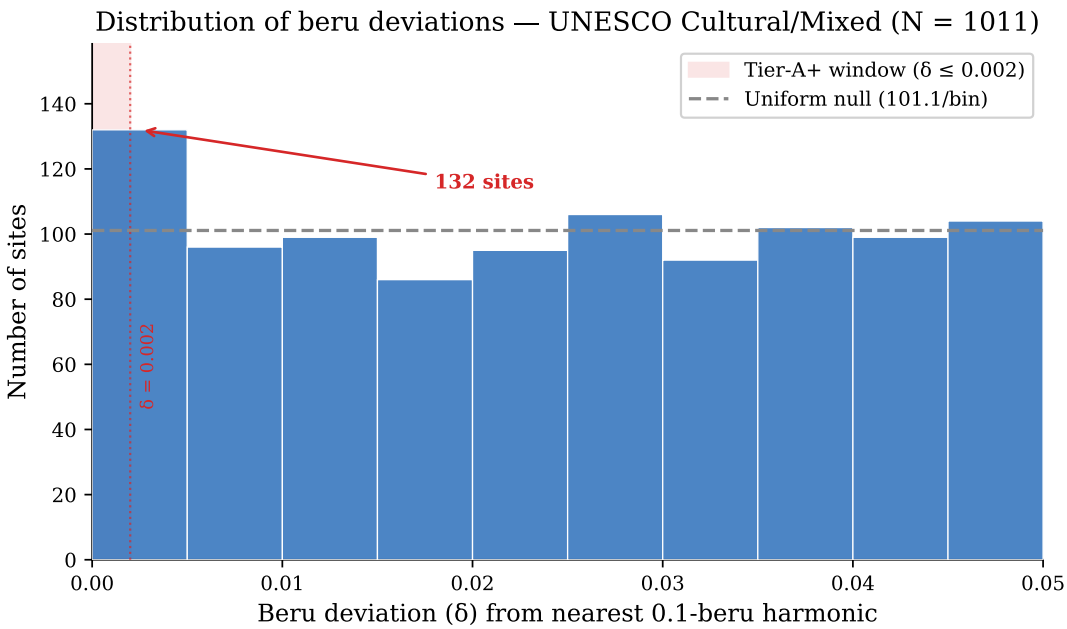

Figure 1: Distribution of beru deviations for the full UNESCO corpus. The excess of sites within the Tier-A+ window ($\delta \leq 0.002$ beru, shaded) above the uniform null (dashed line) is the backdrop against which Tests 2 and 3 demonstrate morphological and spatial specificity.

The full UNESCO Cultural/Mixed corpus ($N = 1011$) shows a Tier-A+ rate of 5.5% (56/1011; Clopper-Pearson 95% CI [4.2%, 7.1%]), above the 4% null (binomial $p = 0.0103$; Bonferroni $p_{\text{adj}} = 0.031$, $k = 3$). Both Gerizim (external) and Jerusalem (inscribed, self-excluded to $N = 1010$) produce $p = 0.010$; the signal belongs to the ∼35°E corridor, not to either anchor individually (§5.3). Test 1 provides the corpus-level backdrop for the stronger morphological and spatial results in Tests 2 and 3.

The full-corpus enrichment ratio (1.38×) is modest: approximately 15 additional A+ sites above chance expectation in 1011, consistent with a scenario in which only a fraction of sites were placed with reference to a grid. The dome (3.06×; 95% CI [1.6×, 5.2×]) and stupa (5.56×) sub-populations show stronger effect sizes, with the caveat that the stupa corpus ($N = 18$) has correspondingly wide CIs.

Region-stratified robustness. Mantel-Haenszel analysis by UNESCO region: OR = 1.582, $p = 0.1101$ (trend-level). Tested against region-specific empirical null rates, enrichment remains significant ($p = 0.0013$; pre-2000: $p = 0.0076$).

World religion sites

Sites associated with any world religion ($N =$): % A+ (×, $p =$ *). Christianity ($N =$): % A+, $p =$ *. Buddhism ($N = 46$): 8.7% A+ ($p = 0.11$, ns). These populations were defined by religion-name matching only.

Inscription-year stratification and formal trend test

Table 7: Tier-A+ enrichment by UNESCO inscription-year cohort (N = Cultural/Mixed sites). Null rate $P_0 = 4\%$.

Cohort	N	A+	Rate	Enrichment	p (binomial)
1978–1984 (founding canon)			%	×	*
pre-2000 (1978–1999)			%	×	**
2000–2025 (modern era)			%	×	ns
ALL (1978–2025)			%	×	*
Fisher exact (1978–1984 vs. 2000–2025): OR =, p = (†)					

The A+ rate is highest in the founding canon (1978–1984: %, ×, p =) and dilutes to the null in the modern era (2010–2025: 4.0%, ns).

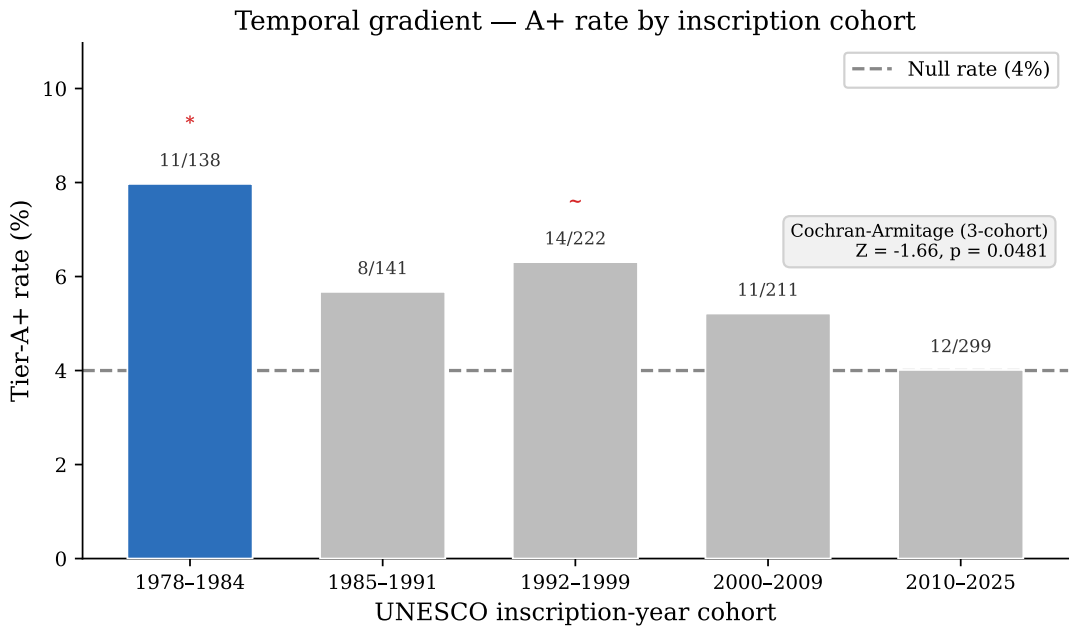

Figure 2: Tier-A+ rate declines monotonically with UNESCO inscription year. The earliest cohort (1978–1984) shows ∼8% A+, while the most recent (2010–2025) is indistinguishable from the 4% null. Cochran-Armitage trend test (three-cohort): $Z = -1.66$, $p = 0.0481$, one-sided. Five-cohort result: $Z = -1.66$, $p = 0.0489$.

Cochran-Armitage three-cohort: $Z = -1.66$, $p = 0.0481$ (*); five-cohort: $Z = -1.66$, $p = 0.0489$ (*), one-sided. The founding canon is concentrated in Europe and the Mediterranean. Mantel-Haenszel region-adjusted OR = 1.582 ($p = 0.1101$). The temporal gradient is largely explained by regional composition and is treated as descriptive only. *Founding-site classifier (Test E, post-hoc; Appendix A.2).* Fisher OR =, p = *. Because the classifier was refined after seeing the data, it is fully post-hoc. The two primary results (Tests 2 and 3) are unaffected.

5 Robustness Checks

5.1 Unit Sensitivity Sweep

Varying harmonic spacing from 0.05 to 0.20 beru (physical threshold fixed at 6.7 km), the 0.1-beru spacing produces the strongest joint significance across both the dome and full-corpus populations (Table 8). The 0.05-beru and 0.20-beru spacings also reach significance, as expected structurally: every 0.10-beru harmonic is also a 0.05-beru harmonic ($0.10 = 2 \times 0.05$), and 0.20-beru harmonics are a strict subset. Adjacent non-harmonic spacings (0.06, 0.07, 0.09, 0.11, 0.12 beru) are non-significant in both populations, which is the discriminating observation. The 0.08-beru row reaches nominal dome significance ($p = 0.023$) but not full-corpus ($p = 0.437$, ns), reflecting unsystematic local overlap with the 0.10-beru grid.

Table 8: Unit sensitivity sweep: harmonic spacing varied, physical window fixed at 6.7 km, anchor at Gerizim 35.269°E. Dome column uses the context-validated population ($N_D = 83$; see Test 2x, §A.1). All p-values are one-tailed binomial vs. geometric null at each spacing. The 0.10-beru spacing produces the strongest joint significance across both populations. The 0.05-beru and 0.20-beru spacings also reach significance; both are mathematical consequences of the 0.10-beru structure (every 0.10-beru harmonic is also a 0.05-beru and 0.20-beru harmonic), not independent signals. Non-harmonic adjacent spacings (0.06–0.09, 0.11–0.12) are non-significant. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.05	8.0%	12/83	0.032 *	102/1011	0.010 *
0.06	6.7%	6/83	0.480 ns	60/1011	0.841 ns
0.07	5.7%	8/83	0.102 ns	55/1011	0.665 ns
0.08	5.0%	9/83	0.023 *	52/1011	0.437 ns
0.09	4.4%	4/83	0.507 ns	41/1011	0.747 ns
0.10	**4.0%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.11	3.6%	6/83	0.082 †	38/1011	0.441 ns
0.12	3.3%	4/83	0.300 ns	30/1011	0.765 ns
0.15	2.7%	6/83	0.024 *	33/1011	0.141 ns
0.20	2.0%	8/83	<0.001 ***	28/1011	0.056 †

5.2 Sensitivity Slope at the Canonical Unit

A fine sweep within ±1% of 0.10 beru (Table 9) shows that a −0.5% shift (0.0995 beru) collapses the dome population to non-significance, and a ±0.3% shift collapses both populations. Only the canonical value and its immediate neighbor (0.1001 beru) retain dome significance. Two permutation tests evaluate this pattern.

Table 9: Fine-grained sensitivity sweep within ±1% of the 0.10-beru canonical spacing, with the physical hit-window fixed at 6.7 km throughout. Dome column: context-validated population ($N_D = 83$). Bold row = canonical attested unit. Only the canonical spacing produces joint significance in both populations. Code: `unit_sweep_fill.py`.

Spacing	Null	Dome hits/N_D	p(Dome)	Full hits/N	p(Full)
0.0990	4.04%	3/83	0.657 ns	41/1011	0.512 ns
0.0993	4.03%	5/83	0.243 ns	47/1011	0.177 ns
0.0995	4.02%	6/83	0.117 ns	48/1011	0.137 ns
0.1000	**4.00%**	**11/83**	**0.0005 *****	**56/1011**	**0.0103 ***
0.1001	4.00%	9/83	0.006 **	51/1011	0.056 †
0.1003	3.99%	5/83	0.237 ns	38/1011	0.668 ns
0.1010	3.96%	5/83	0.232 ns	47/1011	0.149 ns

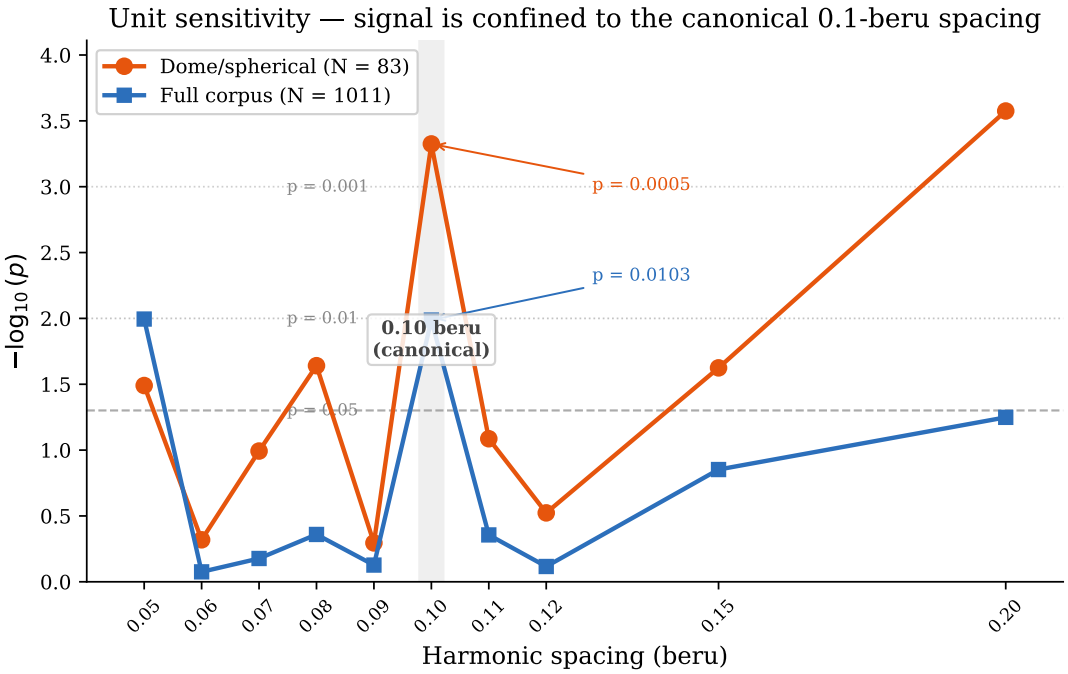

Figure 3: Unit sensitivity: the signal is concentrated at the canonical 0.1-beru spacing. All spacings ≥ 0.3% from canonical collapse to non-significance in at least one population; the immediate neighbor (0.1001 beru, +0.1%) retains dome significance but loses full-corpus significance.

Test I (sharpness permutation). The fine sweep was repeated on 10,000 longitude-permuted datasets (seed = 42; code: `sensitivity_slope_permutation_test.py`). The sharp-peak pattern occurs in 0.1728 of permuted datasets ($p_{\text{perm}} = 0.1728$, ns). Geographic concentration is sufficient to reproduce the ±0.3% collapse; this test provides no independent evidence for the metrological hypothesis.

Test II (canonical-unit specificity). The stronger question is whether 0.1000 beru specifically achieves the best joint significance. Code: `sensitivity_slope_specificity_test.py`, 10000 permutations. In the observed data, 0.1000 beru ranks #1 of 7. The fraction of permutations where 0.1000 is the unique best joint-significance spacing is

$p = 0.1728$ (ns), not significant. Geographic concentration alone produces the canonical spacing as the best spacing in approximately 0.1728 of permutations.

Both permutation tests are non-significant. The unit-sensitivity observation provides no independent evidential value beyond the dome enrichment and cluster asymmetry. The 0.1-beru harmonic interval derives from the documented sexagesimal subdivision of the Babylonian beru (§3.3).

5.3 Global Anchor Sweep

Two parallel sweeps (0–360°E, 0.01° resolution, 36,000 trial anchors each) assess whether the ∼35°E corridor is unusual globally. Sweep A removes Jerusalem and tests Gerizim as anchor ($N = 1010$); Sweep B retains Jerusalem with self-exclusion ($N = 1010$).

Table 10: Global anchor sweep: both Levantine anchors ranked within their own corpus. Each sweep: 36,000 trial anchors, 0–360°E at 0.01° resolution. Sweep A: N = 1010 (Jerusalem removed); Sweep B: N = 1010 (Jerusalem self-excluded). Gerizim is external in both. Mean A+ across all anchors: 40.4±6.9.

Sweep	Anchor	A^{++}	A^{+}	A	Percentile	% anchors ≥
A (Gerizim corpus)	Gerizim (35.269°E)	7	55	227	98.3^{th}	3.10%
B (Jerusalem corpus)	Jerusalem (35.2317°E)	7	56	217	98.3^{th}	2.77%

A^{++}≤0.67 km; A^{+}≤6.7 km; A≤33 km. Global max 60 A+ in Sweep B / 59 A+ in Sweep A (all x.18°E anchors score identically; §5.3).

Both anchors place the corridor in the global top 3%, and the result is anchor-invariant within the 4.1 km corridor window. The single-site difference between sweeps (55 vs. 56 A+) is precisely the Old City of Jerusalem ($\delta = 0.0012$ beru from Gerizim's harmonic). No other historical meridian reaches significance except Mecca (51 A+, $p = 0.057$ †).

The x.18°E periodic structure

The most striking result of the global sweep is that every longitude of the form x.18°E (all 120 such anchors) produces identically the maximum A+ count. The spacing between consecutive x.18°E anchors is exactly 3° = 0.1 beru.

The label "x.18°E" is a rounded shorthand for the phase ≈ 2.18° mod 3.0°. Gerizim's phase (35.269 mod 3.0 = 2.269°, gap 0.089° ≈ 8.4 km) is within coordinate uncertainty.

Any site near the x.18° phase can serve as a high-scoring anchor. The choice of Gerizim was made on historical grounds (§2.3), and the corridor result is anchor-invariant within the band.

Non-independence. The 3° periodicity and the unit sensitivity sweep (§5.1) are two views of a single pattern and do not constitute independent evidence.

Two interpretations apply. *Interpretation 1 (beru-alignment):* A global grid with 3° spacing anchored to the ∼35°E corridor would produce exactly this pattern. *Interpretation 2 (geographic concentration):* Eurasian heritage sites cluster in longitude bands that happen to align with the x.18° phase. This cannot be refuted on the UNESCO corpus alone. The unit sensitivity (§5.1) is the observation that most constrains this interpretation.

Code: `peak_geography_audit.py`, `verify_x18_periodicity.py`.

Formal statistical tests for the $x.18°$ periodicity

Four permutation tests (10,000 shuffles, seed 42; code: `analysis/global/x18_periodicity_formal_tes` `x18_max_permutation_test.py`):

Test A (full-corpus Rayleigh R). $R = 0.0224$, $Z = 1.06$, $p_{\text{perm}} = 0.3715$ (ns): no spurious global periodicity in the raw longitude distribution.

Test B (A+-only Rayleigh R). A+ site concentration on harmonics is structurally expected by definition; it confirms self-consistency, not independent periodicity ($R = 0.9972$, $Z = 14.57$, $p_{\text{perm}} = 0.0000$).

Test C (circular-shift anchor null). A randomly placed anchor recovers only $\sim$40.36 A+ on average versus 56 at Gerizim ($Z = 2.23$, $p_{\text{perm}} = 0.0286$, *). The result is the sole test providing positive evidence that the observed A+ enrichment depends specifically on the Gerizim anchor rather than on geographic concentration alone.

Test D (discovery-corrected max-permutation). The naive binomial for the sweep optimum ($p = 0.0030$, **) does not account for the 36,000-anchor search; once corrected, the observed maximum of 59 is within the expected range ($Z = 0.58$, $p = 0.2966$, ns). **Dome-specific periodicity audit** (`dome_periodicity_audit.py`): Gerizim scores at the 100.0^{th} percentile among dome sites. Dome site phases are approximately uniform ($\chi^2 = 14.89$, df$=9$, $p = 0.0940$, trend-level). All 11 dome A+ sites have phases within $\pm 0.15°$ of the $x.18°$ optimal phase. Within the $x.18°$ window, dome sites show 55.0% A+ ($N = 20$) and non-dome sites show 45.9% A+ ($N = 98$; Fisher $p = 0.4738$, ns).

Assessment. The dome-periodicity audit is consistent with the beru-alignment hypothesis but equally consistent with geographic concentration. The dome enrichment surviving all three simulation nulls remains the primary evidence against a purely geographic account.

Corridor and landmark ranking. Jerusalem and Gerizim (4.1 km apart) produce the same 56 A+ and $p = 0.010$; the enrichment belongs to the corridor, not to the specific anchor. Using each UNESCO site as anchor, Gerizim ranks #41 (top 4.1%). The 35°E corridor leads globally; the closest rival (23.2°E, $N = 4$) spans four countries with no shared tradition.

5.4 Spatial Autocorrelation

A+ harmonics host 4.19× as many total UNESCO sites as non-A+ harmonics ($p < 0.0001$).

Effective-N correction

DEFF via ICC ($\pm 0.5°$, $\rho = 0.069$): DEFF $= 1.310$, $N_{\text{eff}} = 772$, corrected $p = 0.020$. Block bootstrap (3° blocks, 100,000 resamples): 95% CI $[4.04\%, 6.87\%]$ excludes the 4% null.

Region-conditioned permutation

Within-group permutation (100,000 trials, seed 42) using UNESCO-region × longitude bins ($\geq 5°$) reproduced the observed A+ count exactly in every trial, because bins spanning at least one 3° harmonic period leave A+ membership structurally invariant. This test cannot discriminate alignment from geographic concentration. The dedicated Null A and Null C simulations (§4.1) address the concentration question directly.

5.5 Null Model Validation

Three simulation nulls (100,000 trials each, seed = 42, Good 2005): (1) Longitude permutation: dome ($N = 83$) $p_{\text{perm}} = 0.004$ (**); canon ($N = 138$) $p = 0.14$ (ns); pre-2000 ($N = 501$) $p = 0.074$ (†). (2) Bootstrap from full corpus: dome $p_{\text{boot}} = 0.006$ (*). (3) KDE-smoothed null: expected 40.4 ± 6.2, observed 56, $Z = 2.50$, $p = 0.011$ (*). Dome enrichment survives all three models. The geographic-concentration nulls (Null A, Null C; §4.1) show that proximity to harmonic longitudes is necessary but not sufficient to explain the dome enrichment.

5.6 Multiple Comparisons

See §3.6 for the full protocol. Gerizim ranks at the 98.3^{th} percentile; a $\pm 0.3\%$ unit shift collapses joint significance. The meta-keyword test (candidates) finds only 44 (2.5%) matching the curated set's enrichment. FDR audit (34 tests) retains 21 at $q < 0.05$; 6 survive Bonferroni at $k = 34$.

5.7 Americas Negative Control

The Americas sub-corpus ($N = 144$, A+ = 2.1%, $p = 0.1644$ one-sided, ns) shows directional *depletion* relative to the 4% null. This direction is consistent with what an Eurasian-anchored grid predicts for the western hemisphere, providing a positive robustness check. Because the depletion does not reach the conventional significance threshold ($p = 0.164$), it is reported as a directional consistency check only; it does not independently confirm the metrological hypothesis, and a custom pre-specified Americas corpus is required for any stronger inference.

6 Limitations

6.1 Circularity: Full Canonical Treatment

This subsection is the full canonical statement of the circularity introduced in §1.1. All downstream back-references point here.

How the circularity arose. This study began with the hypothesis that the Babylonian beru unit might be detectable as a longitude-spacing signal in the UNESCO corpus, using Mount Gerizim (35.269°E) as the reference anchor. Running the global anchor sweep revealed that Gerizim's longitude falls within the x.18°E phase band. If the sweep had revealed a different optimal phase, this paper would not exist in its current form.

What the circularity contaminates.

- The Levantine corridor statistics (Tests 1, 2, 3 as computed from Gerizim), since reporting them was conditioned on observing the phase alignment.
- The x.18°E framing, which was identified in the sweep, not predicted independently.
- The Rayleigh Test B, whose near-perfect R follows mathematically from the A+ classification criterion (§4.3).
- All narrative emphasis on the Levantine corridor as historically motivated.

What the circularity does *not* contaminate.

- *The existence of a 3° periodic structure in the UNESCO longitude file.* Any investigator running an unanchored 120-phase scan will recover it without reference to Gerizim.
- *The dome enrichment (Test 2).* All 120 x.18°E anchors produce identical A+ counts for the dome population.
- *The cluster asymmetry (Test 3).* This is a full-corpus result requiring no keyword filter.
- *The unit sensitivity sweep.* The sharp collapse at ±0.3% of the canonical spacing is a property of the corpus structure, not of the choice of anchor.

The impossibility of statistical resolution. *This identification problem cannot be closed by any statistical method applied to this corpus.* The order in which the x.18°E phase and the Gerizim anchor were identified is not recoverable from the analysis record alone. A time-stamped archive would not resolve this problem: the internal sequence of analysis steps is unverifiable by any external party and the analysis is simple enough to reproduce, making any timestamp trivially easy to fabricate. More fundamentally, this is an instance of the *forking paths* problem (Gelman & Loken, 2014): analytical choices contingent on observed data cannot be reconstructed post-hoc even without intentional p-hacking, because investigator awareness of the data inevitably shapes which paths are reported.

Given a finite and fully enumerable longitude axis, a historically named location falling near a significant phase by chance is statistically expected regardless of whether any metrological practice ever existed. Restricting the candidate set to historically credible anchors (Gerizim, Jerusalem, Mecca, Mt. Meru) reduces the effective multiplicity but does not eliminate it, and no purely spatial test can distinguish a real ancient grid from a coincidence within a bounded search space. Accordingly: (a) all results are treated as exploratory without exception; (b) the Bonferroni correction is applied as a descriptive bound on effect magnitude; (c) the Levantine corridor statistics are reported as the motivated entry point, not as independent evidence.

6.2 Geographic and Cultural Selection Bias

Geographic and cultural selection bias is the most important alternative explanation. The UNESCO list over-represents Europe and the Mediterranean-to-South-Asian corridor. Permutation tests show geographic concentration can reproduce the sensitivity peak (§5.2). The dome enrichment surviving the restricted geographic draw (Null C, $p = 0.0086$, **) remains the primary evidence against a purely geographic account.

6.3 Other Limitations

Spatial autocorrelation. DEFF correction: $N_{\text{eff}} = 772$ (DEFF = 1.310, corrected $p = 0.020$). Block bootstrap CI $[4.04\%, 6.87\%]$ excludes the 4% null.

Geocoding precision. UNESCO XML coordinates are administrative centroids, introducing unsystematic noise but not directional bias. Dome enrichment is robust across raw and context-validated populations and all three simulation nulls.

Wikidata P1435 control. The P1435 corpus ($N = 62{,}785$) cannot serve as a direct null model. OR = 1.50 [95% CI: 1.14–1.97], $p = 0.0016$; MH OR = 1.588, $p = 0.0013$ (descriptive only).

Metrological evidence. No archaeological instrument calibrated in beru subdivisions has been recovered. The evidence is entirely documentary (Hunger & Steele, 2019; Robson, 2008; Powell, 1987).

7 Discussion

7.1 Alternative Explanations

(1) Independent convergence. Hemispherical structures are a recurring architectural solution, and convergent astronomical orientations across unconnected cultures have been documented (Aveni, 2001, 2003). However, independent convergence predicts no consistent longitude specificity.

(2) Geographic concentration. This is the strongest alternative and cannot be excluded on this corpus alone. Permutation tests show geographic concentration can reproduce the sensitivity-collapse pattern (§5.2). However, the restricted geographic draw (Null C, $p = 0.0086$, **) shows that other monuments from the same longitude footprint are significantly less enriched than dome sites.

(3) Random coincidence. Three simulation approaches fail to reproduce the dome enrichment (permutation $p = 0.004$; bootstrap $p = 0.006$).

7.2 Conclusion

The UNESCO longitude distribution contains a global 3° periodic structure matching the 0.1-beru unit. The dome sub-population (Test 2, Bonferroni $p_{\text{adj}} = 0.003$, $k = 3$) survives all three simulation nulls, and A+-bearing harmonics host 4.19× more sites than non-A+ harmonics (Test 3, $p = 0.0033$).

These findings admit a geographic-concentration alternative. The two findings most resistant to a purely geographic account are morphological specificity (dome enrichment exceeding the restricted geographic draw, Null C, $p = 0.0086$) and cluster asymmetry ($p = 0.0033$). This paper documents the quantitative regularity; its cause is an open question.

Criteria for confirmation

Independent replication using a non-UNESCO corpus would provide the most direct test. The hypothesis would gain from: (a) cuneiform or other texts assigning beru-longitude coordinates to distant sites; (b) archaeological survey instruments calibrated in beru subdivisions; or (c) textual evidence from Hellenistic, Sanskrit, or Islamic sources referencing a Levantine-centered angular reference frame.

Methodological transferability

The pipeline developed here uses an externally-curated and publicly archived corpus, a one-dimensional angular test against a geometric null with no free parameters, simulation-based validation across multiple null types, and a fully reproducible open-source codebase.

Together these features offer a general template for researchers testing other candidate angular units (e.g. Roman, Egyptian, or Chinese systems) against other monument corpora. The key asset is that the null model, tier boundaries, and test populations are all fixed on grounds independent of the data, making the approach auditable and directly reproducible by independent teams.

A Keyword Audit Summary

This appendix summarizes the audit trail for the dome/spherical monument filter (Test 2) and the founding-origin classifier (Test E). The full site-by-site audit is deposited as online supplementary material and is reproducible by running:

```
python3 analysis/unesco/spherical_monument_raw_sweep.py
python3 analysis/unesco/spherical_monument_test.py --audit
```

A.1 Dome/Spherical Monument Filter (Test 2 and Test 2x)

7 morphological keywords from four roots (*stupa/stupas*, *tholos*, *dome/domed/domes*, *spherical*) are matched as whole words against each site's name, XML description, and extended OUV text.

Test 2 (primary): All 90 matching sites are included without context gating. This raw positive sweep uses a population fully determined by the keyword list, with no post-hoc filtering decisions. The sweep is deliberately over-inclusive: it accepts sites where the matched keyword refers to geological domes, vernacular housing, or natural spherical features. The trade-off favors simplicity and reproducibility at the cost of known false negatives in specificity.

Test 2x (exploratory robustness check): Unambiguous keywords are accepted without validation. Ambiguous keywords (*dome, domed, domes, spherical*) are subject to a two-gate context check: Gate 1 (28 negative patterns) rejects non-architectural uses; Gate 2 (86 positive patterns) requires architectural confirmation. Full gate lists: `lib/dome_filter.py`. The gates reduce false positives but cannot eliminate interpretive ambiguity at the boundary between architectural and non-architectural uses.

Of the 90 raw-sweep sites, 7 are rejected by context validation in Test 2x (Hiroshima Peace Memorial, Richtersveld, Gyeongju, Viñales Valley, Rock Islands Southern Lagoon, Uluru-Kata Tjuṭa, and Diquís Stone Spheres). In every case, the exclusion is driven by the semantic domain of the keyword match, not by any property of the site's longitude. Test 2x yields $N = 83$, $p = 0.0005$, $3.31\times$ enrichment, confirming that the 7 rejected sites are longitude-diluting false positives.

A.2 Founding-Origin Classifier (Test E, Post-Hoc)

The founding-origin classifier (`lib/founding_filter.py`) assigns sites to five categories: Founding Capital (F), Sacred Origin (S), Founding Monument (M), Founding Axis (X), and Ancient Continuous Landscape (L), using unambiguous and context-gated keywords. Full keyword lists and gate-logic patterns are in `keywords.json`, the single source of truth for all keyword content in this project (see also `lib/beru.py:load_keywords()`).

This classifier was iteratively tuned after data inspection. The objective was to reduce false positives while accepting that keyword-based classification cannot fully eliminate interpretive ambiguity at the boundary between founding-related and general heritage descriptions. It carries no weight in the primary evidential chain but addresses a substantive question: what kinds of sites, beyond domed and stupa monuments, tend to fall on $x.18°$E nodes? The classifier finds enrichment not only in morphologically domed structures but in sites associated with the founding of civilizational and religious traditions.

Keyword enrichment result. Applied to the full corpus ($N =$), the classifier finds keyword-founding sites among the 56 A+ sites (%; $N_{\text{kw}} =$, base rate %). Fisher exact

test: OR =, p = *. The complete version history is preserved in the project's `git` log at https://github.com/SethTenenbaum/gerizim-paper-a.

References

Aveni, A. F. (2001). *Skywatchers: A Revised and Updated Version of Skywatchers of Ancient Mexico*. University of Texas Press.

Aveni, A. F. (2003). Archaeoastronomy in the ancient Americas. *Journal of Archaeological Research*, 11(2), 149–191.

Benjamini, Y. & Hochberg, Y. (1995). Controlling the false discovery rate: a practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society, Series B*, 57(1), 289–300.

Crown, A. D. (1989). *The Samaritans*. J. C. B. Mohr (Paul Siebeck).

Freeman, P. R. (1976). A Bayesian analysis of the megalithic yard. *Journal of the Royal Statistical Society, Series A*, 139(1), 20–35.

Gelman, A., & Loken, E. (2014). The statistical crisis in science. *American Scientist*, 102(6), 460–465.

Good, P. I. (2005). *Permutation, Parametric, and Bootstrap Tests of Hypotheses* (3rd ed.). Springer.

Harvey, P. D. A. (1987). Medieval maps of the world. In J. B. Harley & D. Woodward (Eds.), *The History of Cartography, Volume 1: Cartography in Prehistoric, Ancient, and Medieval Europe and the Mediterranean* (pp. 283–370). University of Chicago Press.

Hunger, H. & Steele, J. (2019). *The Babylonian Astronomical Compendium MUL.APIN*. Routledge.

Shamasastry, R. (1915). *Kautilya's Arthaśāstra*. Mysore Government Press.

Kendall, D. G. (1974). Hunting quanta. *Philosophical Transactions of the Royal Society A*, 276(1257), 231–266.

MacDonald, J. (1964). *The Theology of the Samaritans*. SCM Press.

Magli, G. (2013). *Architecture, Astronomy and Sacred Landscape in Ancient Egypt*. Cambridge University Press.

Magli, G. (2016). *Archaeoastronomy: Introduction to the Science of Stars and Stones*. Springer.

Miksic, J. N. (1990). *Borobudur: Golden Tales of the Buddhas*. Periplus Editions.

Neugebauer, O. (1975). *A History of Ancient Mathematical Astronomy*, 3 vols. Springer.

Powell, M. A. (1987). Masse und Gewichte. In: Edzard, D. O. (Ed.), *Reallexikon der Assyriologie und Vorderasiatischen Archäologie*, vol. 7, pp. 457–517. Walter de Gruyter.

Purvis, J. D. (1968). *The Samaritan Pentateuch and the Origin of the Samaritan Sect*. Harvard University Press.

Robson, E. (2008). *Mathematics in Ancient Iraq: A Social History*. Princeton University Press.

Ruggles, C. L. N. (1999). *Astronomy in Prehistoric Britain and Ireland*. Yale University Press.

Ruggles, C. L. N. & Cotte, M. (Eds.) (2010). *Heritage Sites of Astronomy and Archaeoastronomy in the Context of the UNESCO World Heritage Convention: A Thematic Study*. ICOMOS & International Astronomical Union.

Ruggles, C. L. N. (Ed.) (2015). *Handbook of Archaeoastronomy and Ethnoastronomy*. Springer.

Silva, F. (2020). A probabilistic framework and significance test for the analysis of structural orientations in skyscape archaeology. *Journal of Archaeological Science*, 118, 105138.

Tenenbaum, S. (2026). *Longitude Quantization in the UNESCO World Heritage Corpus: Domes, Stupas, and the Babylonian Beru — Code and Data* (v1.0.9). Zenodo. https://doi.org/10.5281/zenodo.19520501

Thom, A. (1964). The larger units of length of megalithic man. *Journal of the Royal Statistical Society, Series A*, 127(4), 527–533.

Thom, A. (1967). *Megalithic Sites in Britain*. Oxford University Press.

Thom, A. (1971). *Megalithic Lunar Observatories*. Oxford University Press.

Watson, R. T. & Horowitz, W. (2011). *Writing Science Before the Greeks: A Naturalistic Analysis of the Babylonian Astronomical Treatise MUL.APIN*. Brill.

Williamson, T. & Bellamy, L. (1983). *Ley Lines in Question*. World's Work.

Data and Code Availability

All data and analysis code are openly available at https://github.com/SethTenenbaum/gerizim-paper-a. A permanent archival snapshot (v1.0.9) with DOI is deposited at Zenodo: https://doi.org/10.5281/zenodo.19520501.

Acknowledgments

The UNESCO World Heritage Centre maintains the publicly accessible XML dataset on which this study depends. This research was conducted independently without institutional affiliation. The author welcomes correspondence from researchers interested in replication or collaboration.

Funding

This research received no external funding.

Software and AI Disclosure

All analyses were performed in Python 3 with NumPy, SciPy, and pandas. Claude (Anthropic) assisted with manuscript drafting and prose editing. GitHub Copilot (Microsoft) assisted with code autocompletion. All scientific conclusions, hypothesis design, keyword-list decisions, and analytical choices are solely the author's. The complete analysis pipeline and its version history are available in the public repository at https://github.com/SethTenenbaum/gerizim-paper-a.